Tiffany Wing Lun He

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EDUCATION

Swarthmore College — **Swarthmore, PA**
B.A. in Mathematics and Philosophy, Politics, and Economics — *Aug 2026 – Present*

- **GPA:** 4.0/4.0
- **Relevant Courses:** Introduction to Geometry (Advance Algebraic Geometry), Economics of Crime and Social Justice
- **Activities:** Finance Development Society, Inclusive Excellence Fellows Initiative Student Associate, SwatVotes

BASIS International School Park Lane Harbour — **Huizhou, Guangdong, CN**
High School Diploma — *Aug 2022 – May 2026*

- **GPA:** 4.7+ (Weighted), 3.9 (Unweighted); **SAT:** 1600
- **2026 ISEF (International Science and Engineering Fair):** China National Selection First Place; International Greenwich Olympiad Team China Member, International Finalist
- **2024 International Logic Olympiad:** Team China Member, Gold Medalist
- **2024 S.-T. Yau Science Award:** National First Prize (Top 10), Math Category
- **2023 American Mathematics Competitions 10 & 12:** Honor Roll of Distinction (Top 1%), First Place in School, Certificate of Achievement, USAMO & USA(J)MO Qualification; scored 12 on 2024 AIME
- **2023 British Math Olympiad:** National Top 80, invited to BMO China Camp, Best in School
- **2021–2022 China Youth Mathematics Olympics:** National Gold Medal

Stanford University — **Palo Alto, CA**
University-Level Online Courses — *Aug 2023 – May 2025*

- Earned 12 credits in Linear Algebra, Multivariable Calculus, Real Analysis, and Complex Analysis

RESEARCH & PROFESSIONAL EXPERIENCE

Swarthmore College Social Science Quantitative Laboratory
Student Associate & Researcher — *Sep 2026 – Present*

- Independent research with *Prof. Greg Amusu* on developing an open-source redistricting risk engine that models congressional maps as portfolios of correlated bets, using an ecological-inference-corrected Bayesian factor model, importance sampling, risk-tilted Sequential Monte Carlo and Bayesian inverse optimization to estimate seat CVaR and recover mapmakers' objectives, with conformal stress tests and copula-based prediction-market analysis
- Lead student research group on designing a companion system that measures politicians' risk tolerance via a cross-office Bayesian IRT model and structural mixed-logit estimation of career decisions
- Hold weekly R tutoring clinic sessions for 50+ students in econometrics and quantitative research; authored a 60+ page quantitative research guide; lead monthly workshops in quantitative research

Point72 Academy National Case Competition
Team Captain — *Oct 2026*

- Pitched FTAI Aviation with a 12-month target of $244 (+46%) and 6.0x bull/bear reward-to-risk. Used a reverse DCF to show the share price implied market-share growth stalling near 18%, versus our 21% forecast; translated that gap into explicit revenue, margin, and valuation assumptions.
- Reframed engine retirements—the market's key concern—as a partial natural hedge: retirements reduce competition for engine parts and lower FTAI's input costs, cushioning the downside. Stress-tested the thesis and estimated fundamental downside of about 11%.
- Built an integrated Excel/Python valuation model linking operating forecasts, cash flows, and valuation. Ran a 20,000-scenario Monte Carlo to quantify target-price outcomes, tested sensitivities to market share, margins, and valuation multiples, and compared FTAI with peers. Converted the analysis into a catalyst-based trade plan with defined upside, downside, and key thesis checkpoints.

Junfeng Bfs Technology Co., Ltd.
Software Engineering Intern — *Jun 2026 – Aug 2026*

- Collaborated with a 20+ member research team on dynamic bone support facilities for the disabled through weight-shifting structures and AI physical detection technologies
- Completed 200+ test trials on human users; collected, cleaned, and analyzed 10,000 data points; co-authored 80+ pages of experimental reports and design improvement suggestions

Mathematics Research on the Solution to Schubert's Problem and Convex Polyhedra
Independent Research — *May 2024 – Dec 2024*

- Investigated the fundamental intersection problem in topology posed by Hermann Schubert in 1879
- Provided 2 original solutions using Morse theory, cohomology, intersection numbers, and convex polyhedra
- Earned National First Prize in 2024 S.-T. Yau Science Award (10/2,000+); qualified for the international round

Maoge Overseas Consulting
Assistant to the President — *Feb 2026 – Present*

- Managed the full-cycle contract execution and tuition settlement process for 40+ student cases, overseeing fee collection, payment reconciliation, and refund approval

EXTRACURRICULAR ACTIVITIES

Coast to Coast
Selected Mentee — *Aug 2026 – Present*

- Participate in a nonprofit mentorship program that aims to prepare students for successful careers in finance and cultivate the next generation of leaders in the industry
- Attend biweekly sessions in investment banking, long short hedge fund, and accounting

Skills: Financial Modeling, Bloomberg, Excel, Python, R, FactSet, Pitchbook; Preqin, Mandarin & Cantonese (Native), English (Native)